POWER PLANT FUNDAMENTALS

BASIC THERMODYNAMICS

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1.0 INTRODUCTION

In order to acquire a basic understanding of power plant operation, it is necessary to have some understanding of certain concepts in the field of thermodynamics. In the broadest sense of the term, thermodynamics is the physical science that deals with energy and energy transformations. The branch of thermodynamics which is of primary interest to engineers is typically referred to as applied thermodynamics or engineering thermodynamics. It deals with the fundamental design and operational considerations of boilers, turbines, internal combustion engines, air compressors, refrigeration and air conditioning equipment, and other machinery in which energy is exchanged or transferred in order to produce a desired effect. This training module deals with certain thermodynamic concepts that are particularly necessary as a basis for understanding the operation of a power plant. The information given here is introductory in nature and is not in any sense a complete or thorough exploration on the subject of thermodynamics. In so far as possible, verbal descriptions are used rather than mathematical analysis to develop an understanding of the laws and principles of energy exchanges and transformations.

1.1 Energy Transfer Systems

An energy transfer system is a system which is specifically designed to carry out the transfer of energy from one form to another. In the simplest of terms, a power plant is an energy transfer system. The energy transformation of greatest importance in the power plant is the production of mechanical work from thermal energy. Power plants depend largely upon the thermal-to-mechanical energy transformation to spin generators in order to generate electricity. However; in addition to the primary energy transfer flowpath which exists in a power plant, many other energy transformations must take place in order for the power plant equipment and systems to operate. For example, the exchange of energy that occurs within the various coolers and heat exchangers through out the plant, as well as the operation of electric motors, pneumatically actuated valves, and even the pumping of fluid through a network of system piping.

For the purpose of understanding and analyzing their operation, the boiler, turbine-generator and condenser may each be considered as an energy transfer system. Likewise, the feedwater system and the basic steam loop may be considered as energy transfer systems. Thus, the boundaries of a system are established based upon the portion of the total system that is being studied.

Regardless of where the boundaries are established in an energy transfer system, each system contains a working fluid in which energy can be stored and removed. A fluid is any substance that conforms to the shape of its container. It may be either a liquid or a gas. In a power plant, the primary working fluid is typically water or steam. In power plant terminology, the term 'system' is most commonly used to describe a collection of hardware, such as the feedwater system or the main steam system. However; when a particular piping system is studied as energy transfer system, it is important to think of the fluid within the pipes as the system and everything else as this system's surroundings.

2.0 PROPERTIES OF WORKING FLUIDS

The properties of a substance are measurable characteristics used to describe it. Properties are either intensive or extensive. An intensive property is one that is independent of mass. It does not depend on how much of the substance is present. For example, pressure and temperature are intensive properties. An extensive property is one that depends on mass. It depends on how much of the substance is present. For example, volume and weight are extensive properties. Rather than working with extensive properties, it is customary to define intensive properties which can be associated with particular extensive properties. For example, the specific volume is defined as the volume per unit mass; it is an intensive property.

The basic properties of the working fluid in an energy transfer system are fundamental to understanding the operation of such a system. Some of these properties, such as mass, volume, temperature and pressure, are familiar concepts and can be defined directly.

2.1 Mass and Weight

The mass (m) of a substance is the measure of the amount of matter present. In the English system, it has units of pounds-mass (lbm). The weight of a substance is the force exerted by the substance when it is accelerated by gravity. In the English system, it has units of pounds-force (lbf). In virtually all engineering applications, the weight of a substance in pounds-force equals the mass of the substance in pounds-mass.

2.2 Volume

The volume (V) of a substance is the measure of the amount of space it occupies. In the English system, volume is measured in units of cubic feet (ft^3) or cubic inches (in^3). The specific volume (v) of a substance is its total volume (V) divided by its total mass (m); it is its volume per unit mass. Specific volume is measured in units of cubic feet per pound-mass (ft^3/lb_m).

$$v = V \div m$$

where: v = specific volume (ft^3/lb_m)
V = volume (ft^3)
m = mass (lb_m)

2.3 Density

The density (ρ) of a substance is its total mass (m) divided by its total volume V; it is its mass per unit volume. Density is measured in units of pounds-mass per cubic foot (lb_m/ft^3). The density (ρ) of a substance is the reciprocal of its specific volume (v).

$$\rho = m \div V$$

where: ρ = density (lb_m/ft^3)
m = mass (lb_m)
V = volume (ft^3)

2.4 Specific Gravity

Specific gravity (SG) is a measure of the relative density or specific weight of a substance compared to that of water. It is defined as the density or specific weight of the substance divided by the density or specific weight of water. The density of water is approximately 62.4 lb/ft^3. Knowing the specific gravity of a substance provides a quick method of determining if the substance will float or sink when surrounded by water. If the specific gravity of a substance is greater than 1.0, it will sink. If the specific gravity of a substance is less than 1.0, it will float. Specific gravity is a unitless quantity since it is the ratio of two quantities having the same units.

2.5 Pressure

The pressure (P) of a system is caused by the collisions of the molecules of the fluid with the boundaries of the system. Except at a temperature of absolute zero, the molecules of any substance are continually moving. In a solid, the molecules are constrained so that they only vibrate. However; in a liquid or a gas, molecules can move extensively. These molecules are continually colliding with each other and with the walls of their container. As they hit the walls, they exert forces that try to push the walls outward. There are billions of molecules in each cubic inch of a liquid or a gas, and they collide billions of times each second with each square inch of the walls. The forces resulting from all of these collisions add up to cause the pressure exerted by a system on its surroundings.

2.6 Temperature

The temperature (T) of a substance is the measure of the molecular activity of the substance. The higher the temperature of a substance, the greater the movement of the molecules within the substance. When a hot body is placed in contact with a cold body, the molecules of the hot body impart motion to the molecules of the cold body. The temperature of the hot body decreases and that of the cold body increases until the two are equal. The energy that flows between the two bodies because of a difference in temperature is called heat. Temperature is not a measure of heat, it is a measure of the ability to transfer heat. Temperature is a property of a system; heat is a form of energy which is transferred to or from a system.

There are four different temperature scales, (1) the Centigrade scale, (2) the Kelvin scale, (3) the Fahrenheit scale and (4) the Rankine scale. All four scales can be defined in terms of the freezing point and boiling point of water under a standard atmosphere of pressure. The freezing point of water is 0°C (degrees Centigrade), 273.15°K (Kelvins), 32°F (degrees Fahrenheit) and 491.67°R (degrees Rankine). The boiling point of water is 100°C, 373.16°K, 212°F and 671.67°R. In engineering applications, temperature is most often given in degrees Fahrenheit and/or Centigrade (Celsius).

Absolute zero is the lowest temperature theoretically attainable. The zero of the absolute temperature scale or absolute zero is taken as a point at which there is an absence of heat energy or no molecular vibration. Absolute zero on the Fahrenheit scale is -459.67°F; on the Centigrade scale it is -273.15°C; on the Kelvin scale it is 0°K; and on the Rankine scale it is 0°R. The absolute temperature of a substance is its temperature in either degrees Kelvin or degrees Rankine. Absolute temperature is written as T_{abs}.

3.0 ENERGY

Although *energy* has a general meaning to almost everyone, it is not easy to define the word in a completely satisfactory manner. Because energy is so often manifested by the production of work, energy is commonly defined as "the capacity for doing work". However, this is not entirely adequate as a definition, since work is not the only effect that is produced by energy. For example, heat can flow from one body to another without doing any work at all, but the heat must still be considered as energy and the process of heat transfer must be recognized as a process that has produced an effect. A broader definition then, and one which satisfies more of the conditions under which we know energy to exist is; *Energy is the capacity for producing an effect.*

3.1 Forms of Energy

Energy exists in many forms. The four types of energy which exists in the energy transfer cycle of a power plant are; (1) chemical energy, (2) thermal energy, (3) mechanical energy, and (4) electrical energy. *Chemical energy* is produced from the rearrangement of molecular bonds in a chemical reaction, such as occurs when fuel is burned in a boiler or in a combustion turbine. To achieve combustion, oxygen and heat are supplied to the fuel and a chemical reaction takes place. The chemical structure of the fuel is altered and the products that result (ash, carbon, soot) are at a lower energy level than the original composition of the fuel. In going from a high energy level to a lower energy level, energy is given off in the form of heat.

Thermal energy is energy associated with the movement of molecules within a substance with respect to temperature or heat. For example, the heat released from the combustion of fuel causes the movement of the surrounding air molecules to increase rapidly in proportion to the amount of heat or thermal energy released by the flame and absorbed by the air. More heat (thermal energy) results in increased movement of a substances' molecules; conversely, less heat results in less movement.

Mechanical energy is the energy associated with the movement of objects; for example, a rotating turbine wheel. Although an outside force or other form of energy is required to achieve and maintain movement of the turbine wheel, the actual movement of the wheel itself is referred to as mechanical energy.

Electrical energy, the flow of electrons, is produced when a conductor moves through a magnetic field. A generator is used to generate electrical energy; i.e. during it's operation it causes electrons to flow. Light, X-rays, and radio waves are examples of electrical energy associated with particles that are even smaller than atoms.

3.2 Energy of a Working Fluid

The total energy of a working fluid or substance is the sum of the magnitudes of the various forms of energy the substance possesses. Normally, the total amount of energy a body possesses cannot be determined in an absolute sense. It must be measured with respect to some reference. Thus, the potential energy of a pound of water is defined in terms of its position above some reference position. This approach is satisfactory for most engineering applications since the change in energy is what relates to the work done.

With regard to thermodynamics, it is common to define energy in terms of the work done by or on a system. The concept that a system possesses a certain quantity of energy which is decreased when the system performs work and increased when work is done on the system is fundamental to understanding energy transfer systems. When the working fluid of a system performs work, its energy is decreased. When work is done on the working fluid, its energy is increased.

There are four forms of energy possessed by the working fluid in an energy transfer system which are important in the analysis of such systems; (1) potential energy, (2) kinetic energy, (3) internal energy and (4) pressure-volume energy.

3.2.1 Potential and Kinetic Energy

Potential energy is stored in a system by virtue of the relative positions of the bodies that make up the system. The potential energy associated with the gravitational attraction between the earth and another body provides us with many everyday examples. A rock resting on the edge of a cliff in such a position that it will fall freely if pushed; has potential energy. Water at the top of a dam has potential energy. A sled that is being held at the top of an icy hill has potential energy. Note that in each of these examples, the energy resides neither in the earth alone nor in the other objects, but rather in an energy system of which the earth is merely one component.

Kinetic energy is stored in a system by virtue of the relative velocities of the component parts of the system. Push the rock over the edge of the cliff, open the gate of the dam, or let go of the sled and something will move. The rock will fall, the water will flow, and the sled will slide down the hill. In each case, potential energy will be changed to kinetic energy. Since it is customary to ascribe zero velocity to an object which is at rest with respect to the earth, it is also customary to think of kinetic energy as though it pertained only to the object which is in motion with respect to the earth. It should be remembered however; that kinetic energy, like potential energy, is properly assigned to the system rather than to any one component of the system.

In the above examples of potential and kinetic energy, an external source of energy was used to get things started. Energy from some outside source is required to push the rock, open the gate of the dam, or let go of the sled. All real machines and processes require this kind of a boost from an energy source outside of the system. Similarly, the energy from any one system is bound to affect other energy systems, since no one system can be completely isolated as far as energy is concerned. From a practical standpoint, it is easier to understand the basic energy concepts if we disregard all the other energy systems that might be involved in or affected by each energy process.

Potential energy and kinetic energy are mutually convertible. To take the example of the rock resting on the edge of the cliff, let us suppose that some external force pushes the rock over the edge so that it falls. As the rock falls, the system loses potential energy but gains kinetic energy. By the time the rock,

reaches the ground at the base of the cliff, all the potential energy of the system has been converted into kinetic energy. The sum of the potential energy and the kinetic energy is identical at each point along the line of fall, but the proportions of potential energy and kinetic energy are constantly changing as the rock falls.

To take another example, consider a baseball that is thrown straight up into the air. The ball has kinetic energy while it is in upward motion, but the amount of kinetic energy is decreasing and the amount of potential energy is increasing as the ball travels upward. When the ball has just reached its uppermost position, before it starts to fall back toward the earth, it has only potential energy. Then, as the ball falls back toward the earth, the potential energy is converted into kinetic energy again.

3.2.2 Internal Energy

Potential energy and kinetic energy are 'macroscopic' forms of energy. In other words, potential and kinetic energy can be visualized in terms of the position and the velocity of the total system. In addition to macroscopic forms of energy, a substance also possesses several 'microscopic' forms of energy. The microscopic forms of energy of a substance are not visible; they consists of the rotation, vibration, and movement among the molecules of the substance. Accordingly, the total sum of the molecular movement within a substance is defined as the *internal energy* of the substance.

At temperatures above absolute zero, the molecules of a substance are constantly rotating, vibrating, or moving from place to place. As more and more heat is added to a substance, the molecular movement within the substance increases; and the internal energy of the substance increases proportionately. The more internal energy a substance has, the faster it's molecules will move. Therefore, it can be concluded that temperature is a relative measure of a substance's internal energy. In singular applications, internal energy is typically represented by the symbol U. In engineering applications, internal energy is expressed in British Thermal Units (BTU).

3.2.3 Pressure-Volume Energy

In addition to internal energy, there is another form of energy which the working fluid of an energy transfer system possesses that is important in understanding the thermodynamics of such a system. This form of energy is defined as *pressure-volume energy* because it arises from the pressure (P) and the volume (V) of the system. Pressure-volume energy is numerically equal to PV; the product of pressure and volume.

Energy can be defined as the capacity of a system to perform work. If a system with pressure and volume is permitted to expand, it performs work on its surroundings. For example, when heat is applied to an enclosed volume of gas, the pressure of the gas will increase. The pressure increase is caused by the increased movement of the molecules. This increased pressure is *pressure-volume energy*. Further heating of the gas causes it to expand even more. As this occurs, thermal energy from the heat source is transferred to the gas where it is stored as both internal energy and pressure-volume energy. Figure 1 illustrates this concept.

Energy from the burning candles is being transferred to the gas inside the container. The candles are generating energy from the combustion of paraffin wax and ambient air. The gas within the sealed container absorbs this energy in two forms; (1) the internal energy of the gas increases as seen by the temperature increase, and (2) the pressure-volume energy increases as seen by the pressure increase based on the fact that the volume of the container does not change.

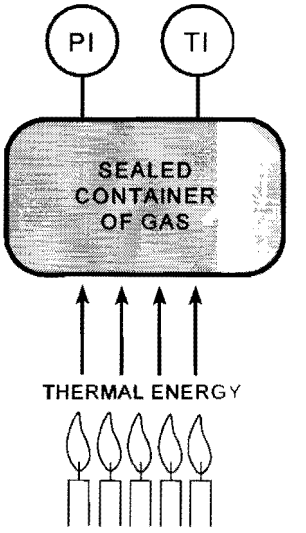

Figure 1 - PV Energy

4.0 WORK

Work is mechanical energy in transition; that is, it is a transitory form of mechanical energy which occurs only between two or more other forms of energy. *Work is done when a tangible body or substance is moved through a tangible distance by the action of a tangible force*. Thus, we may define work as the energy which is transferred by the action of a force through a distance. In the case of work done against gravity, the force is numerically the same as the weight of the object or body that is being displaced.

4.1 Performing Work

As previously illustrated in Figure 1, the release of thermal energy from the burning candles was transferred to a volume of gas sealed within a container. The gas absorbed and stored the thermal energy in the form of internal energy and pressure-volume energy. Taking this process one step further, we can use this stored energy to perform work. Refer to Figure 2 below.

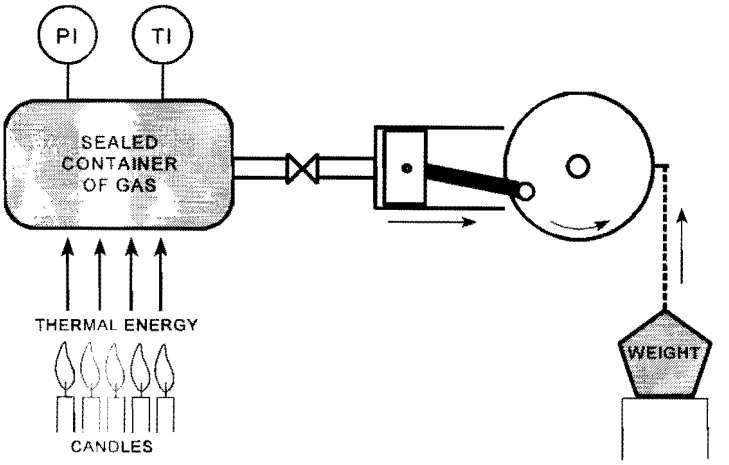

Figure 2 - Stored Energy Doing Work

In the illustration above, a specific amount of work is required to lift the weight off it's pedestal; i.e. moving a tangible body through a tangible distance by the action of a tangible force equals work. By connecting the weight to a flywheel which is driven by a piston/cylinder arrangement, it is possible to lift the weight by utilizing the internal and pressure-volume energy of the gas stored in the container. If the valve connecting the container and the cylinder were opened, the p-v energy of the gas would act upon the piston, forcing it downward into the cylinder. This force would result in a counterclockwise rotation of the flywheel, and as a result, the weight would be lifted.

In the above example, the gas inside the container is under pressure. Pressure is produced by the gas molecules colliding with the walls of it's confined container. When the valve is opened, the pressure does work by exerting a force on the piston, which causes the piston to move. As the gas expands and moves the piston, the volume of the gas increases with a corresponding decrease in pressure. As a result, some of the pressure-volume energy is used to move the piston.

The same principle applies to the use of steam (or gas) within a power plant to generate electricity. Steam flows through the system piping enroute to the turbine because it is trying to expand toward a lower pressure area; i.e from the outlet of the boiler drum to the exhaust end of the low pressure steam turbine. The pressure drops very little before the steam reaches the turbine because the boiler is continually making more steam and very little energy is taken out of the steam in the pipes, just a small amount of friction and heat losses. In the turbine however, the pressure drop is great because the turbine takes the energy out of the steam in the form of mechanical energy as the steam expands towards the lower stages.

Another example of this 'energy conversion' principle is identified in the operation of a motor driven pump and how work is done on a fluid. An electric motor receives electrical energy and converts it into a continuous movement of rotational mechanical energy. Within the pump, the continuous input of rotational mechanical energy is used to perform work. The result is fluid flow and pressure at the discharge of the pump. However; since a liquid is not compressible, it cannot have pressure-volume energy. Instead, the increase in pressure resulting from a pump is normally thought of as mechanical (kinetic) energy. In a fluid system, pressure is the result of the mechanical work that the pump performs on the fluid and the resistant against which the fluid acts on the boundaries of the system. This creation of pressure results in fluid flow.

4.2 Measuring Work

The energy required to move objects is called mechanical energy. Work is actually energy in motion and thus, doing work requires energy. Therefore, work is a method of converting one type of energy to another. It is important to note however, that no work is done unless something is displaced from it's previous position. When a five pound weight is lifted from the floor to a table that is three feet high, 15 foot pounds of work has been done.

Suppose that an object is moved in such a way that it returns to its original position. Has work been done or not? Consider again, the example of lifting a 5 pound weight to the top of a 3 foot table. By this action, 15 foot-pounds of work has been performed. Now suppose that the weight falls back to the floor. Displacement is zero, so work must be zero. But what has happened to the 15 foot-pounds of work that was put into the system when the weight was lifted to the top of the table? By doing this work, the system received 15 foot-pounds of mechanical potential energy. When the weight fell back to the floor, the mechanical potential energy was converted into mechanical kinetic energy.

In one sense, is can be said that the above work was undone and that no 'net work' was performed. On the other hand, one may choose to regard the two actions separately. In such a case, 15 foot-pounds of work was done by lifting the weight, and the force of gravity acting upon the weight has done 15 foot-pounds of work to return the weight to its original position on the floor. In other words, one action of work may be regarded as positive and the other as negative. The two cancel each other out, so there is again no net work. But in this case, it can be recognized that 15 foot-pounds of work were performed twice, in two separate operations, by two different forces.

The above example was elaborated somewhat because several important deductions may be drawn from it. First, it may help to clarify the concept of work as a form of energy that must be accounted for. Also, it may help to convey the real meaning of the statement that *work is mechanical energy in transition*. Work is energy in transition because it occurs only temporarily between other forms of energy, and because it must always begin and end as stored energy. Finally, the example suggests the need for arbitrary reference planes in connection with the measurement of potential energy, kinetic energy, and work. The quantitative consideration of any form of energy requires a reference which defines the starting point and the stopping point of any particular operation; the reference planes are practically always relative rather than absolute.

Note that mechanical potential energy, mechanical kinetic energy, and work are all measured in the same unit, the *foot-pound*. One foot-pound of work is done when a force of 1 pound acts through a distance of 1 foot. One foot-pound of mechanical kinetic energy or 1 foot-pound of mechanical potential energy is the amount of energy that would be required to accomplish 1 foot-pound of work.

The amount of work done has nothing to do with the length of time required to do it. If a weight of 1 pound is lifted through a distance of 1 foot, 1 foot-pound of work has been done, regardless of whether it was done in half a second or half an hour. The rate at which work is done is called *power*. In the field of mechanical engineering, the horsepower (hp) is the common unit of measurement for power. By definition, 1 horsepower is equal to 33,000 foot-pounds of work per minute or 550 foot-pounds of work per second. Thus, a machine capable of doing 550 foot-pounds of work per second is said to be a 1 horsepower machine.

5.0 THERMAL ENERGY

Energy associated primarily with systems of molecules is called thermal energy. Like other kinds of energy, thermal energy may exist in stored form (in which case it is called internal energy) or as energy in transition (in which case it is called heat).

In common usage, the term heat is often used to include all forms of thermal energy. However, this lack of distinction between heat and the stored forms of thermal energy can lead to serious confusion. Therefore; from this point forward, the term internal energy is used to describe the stored forms of thermal energy, and the term heat is used only to describe thermal energy in transition.

5.1 Internal Energy

Internal energy was previously discussed in Section 3.2.2 and was defined as the total sum of molecular movement within a substance with respect to temperature. The movement of molecules within a substance is directly related to the amount of thermal energy contained in the substance.

The magnitude of internal energy is dependent upon the mass of the molecules and the average distance by which they are separated, in much the same way that mechanical potential energy depends upon the mass of the bodies in the system and the distance by which they are separated. The force of attraction between molecules is greatest in solids, less in liquids and yielding substances, and least of all in gases and vapors. Whenever something happens to change the average distance between the molecules of a substance, there is a corresponding change in the internal energy of the substance.

5.2 Heat

Although the term heat is more familiar than the term internal energy, it may be more difficult to arrive at an accurate definition of heat. *Heat is thermal energy in transition*. Like work, heat is a transitory energy form existing between two or more other forms of energy.

Since the flow of thermal energy can occur only when there is a temperature difference between two objects or regions, it is apparent that heat is not a property or attribute of any one object or substance. If a person accidentally touches a hot stove, he may understandably feel that heat is a property of the stove. More accurately however, he might reflect that his hand and the stove constitute an energy system and that thermal energy traveled from the stove to his hand because the stove was at a higher temperature than his hand.

As another example of the difference between heat and internal energy, consider two equal lengths of piping, made of identical materials and containing steam at the same pressure and temperature. One pipe is well insulated, one is not. From everyday experience, we expect more heat to flow from the uninsulated section of pipe than from the insulated section. When the two pipes are first filled with steam, the steam in one pipe contains exactly as much internal energy as the steam in the other pipe. We know this true because the two pipes contain equal volumes of steam at equal pressures and temperatures. After a few minutes, the steam in the uninsulated pipe will contain much less internal energy than the steam in the insulated pipe, as can be indicated by reading the pressure and temperature gages on each pipe. What has happened? Stored thermal energy (internal energy) has moved from one place to another; first from the steam to the pipe, then from the uninsulated pipe to the air. It is this movement, or this flow, of energy that should properly be called *heat*.

To summarize; temperature is a reflection of the amount of internal energy possessed by an object or a substance, and it is therefore an attribute or property of the substance. The movement or flow of thermal energy, or in other words heat, is an attribute of the energy system rather than of any one component of it.

5.2.1 Units of Measurement

In engineering, heat is commonly measured in the unit called the British Thermal Unit (Btu). Originally, one Btu was defined as the quantity of heat required to raise the temperature of 1 pound of water through 1 degree on the Fahrenheit scale. A similar unit called the calorie (cal) was originally defined as the quantity of heat required to raise the temperature of 1 gram of water through 1 degree on the Celsius scale. The Btu and the calorie are also defined in terms of the unit of energy called the joule. The following relationships have thus been established by definition or derived from the established definitions:

1 calorie = 1/860 watt-hour
= 4.18605 joules
= 3.0883 foot-pounds

1 Btu = 251.996 calories
= 1054.886 joules
= 778.26 foot-pounds

5.3 Heat Transfer

Heat transfer is the flow of energy from one body, substance, or region to another. In order for heat transfer to take place, there must be a difference in temperature; i.e., one substance must be warmer than the other. The transfer of thermal energy always flows from a region of higher temperature to a region of lower temperature. In thermodynamics, the higher temperature region may be called the source or the emitting region; the lower temperature region may be called the receiver, or the receiving region.

If a steel block of 400°F is placed next to a steel block of 100°F, energy from the hotter block will travel to the colder block; i.e., one block cools and the other warms. In time, both blocks will be at the same temperature. How fast heat is transferred between the blocks is called the rate of heat transfer. The greater the temperature difference between the blocks, the faster the rate of heat transfer will be. As the blocks approach the same temperature, the rate of heat transfer decreases. When the blocks reach the same temperature, there is no heat transfer.

Heat flows from a warmer substance to a cooler one by three different methods; (1) conduction, (2) convection, and (3) radiation. All three methods of heat transfer take place in a power plant.

5.3.1 Conduction

Conduction is the mode by which heat flows from a hotter to a colder region when there is physical contact between the two regions. For example, consider a metal bar which is held so that one end of it is in boiling water. In a very short time the end of the bar which is not in the boiling water will have become hot also. This occurs because heat has been conducted from molecule to molecule along the entire length of the bar. The molecules in the layer nearest the source of heat become increasingly active as they receive thermal energy. Since each layer of molecules is bound to the adjacent layers by cohesive forces, the motion is passed on to the next layer, which in turn sets up increased activity in the next layer. The process of conduction continues as long as there is a temperature difference between the two ends of the bar.

The total quantity of heat conducted depends upon a number of factors. To explain this; consider a bar of carbon steel which is uniform in cross-sectional area throughout its length. One end of the bar is kept at a uniformly high temperature, the other end is kept at a uniformly low temperature. After a steady and uniform flow of heat has been established, the total quantity of heat that will be conducted through the bar depends upon the following relationships:

1. The total quantity of heat passing through the conductor in a given length of time is directly proportional to the cross-sectional area of the conductor.
2. The total quantity of heat passing through the conductor in a given length of time is proportional to the thermal gradient; that is, to the difference in temperature between the two ends of the bar, divided by the length of the bar.

3. The quantity of heat is directly proportional to the time of heat flow.
4. The quantity of heat depends upon the thermal conductivity of the material of which the bar is made. Thermal conductivity is different for each material. For example, aluminum transfers heat more readily than steel and therefore has a higher thermal conductivity.

5.3.2 Radiation

Radiation is a mode of heat transfer that does not involve any physical contact between the emitting region and the receiving region. A person sitting near a hot stove is warmed by thermal radiation from the stove, even though the air in between remains relatively cold. Thermal radiation from the sun warms the earth without warming the space through which it passes. Thermal radiation passes through any transparent substance (air, glass, ice) without warming it to any extent because transparent materials are very poor absorbers of radiant energy.

All substances (solids, liquids, and gases) having a temperature above absolute zero emit radiant energy at all times. We tend to think of radiant energy as something that is emitted only by extremely hot objects such as the sun, a stove, or a furnace, but this is a very limited view of the nature of radiant energy. A stove radiates energy to everything surrounding it, but at the same time all the surrounding objects are radiating energy to the stove. A child standing near a snowman may well believe that the snowman is "radiating cold" rather than emitting radiant energy. Actually, however, both the child and the snowman are emitting radiant energy. The child of course is radiating far more energy than the snowman, so the net effect of this energy exchange is that the snowman grows warmer and the child grows colder. We are literally surrounded by, and a part of, such energy exchanges at all times.

Thermal radiation is an electromagnetic wave phenomenon, differing from light, radio waves, and other electromagnetic phenomenon merely in the wavelengths involved. When the wavelengths are in the infrared part of the electromagnetic spectrum; that is, when they are just below the range of visible light waves, the radiated energy is referred to as thermal radiation. It should be noted however, that all electromagnetic waves transport energy which can be absorbed by matter and which can in many cases result in observable thermal effects. For example, one energy unit of light absorbed by a substance produces the same temperature rise in that substance as is produced by the absorption of an equal amount of thermal (infrared) energy.

When radiant energy falls upon a body that can absorb it, some of the energy is absorbed and some is reflected. The amount absorbed and the amount reflected depend in large part upon the surface of the receiving body. Dark, opaque bodies absorb more thermal radiation than shiny, bright, white, or polished bodies. Shiny, bright, white, or polished bodies reflect more thermal radiation than dark, opaque bodies. Good radiators are also good absorbers and poor radiators are poor absorbers. In general, good reflectors are poor radiators and poor absorbers.

5.3.3 Convection

Although convection is often loosely classified as a mode of heat transfer, it is more accurately regarded as the mechanical transportation of a mass of fluid (liquid or gas) from one place to another. In the process of this transportation, all the thermal energy stored within the fluid remains in stored form unless it is transferred by radiation or by conduction. Since convection does not involve thermal energy in transition, we cannot in the most fundamental sense regard it as a mode of heat transfer.

Convection is the transportation or the movement of some portions within a mass of fluid. As this movement occurs, the moving portions of the fluid transport their contained thermal energy to other parts of the fluid. The effect of convection is thus to mix the various portions of the fluid. The part that was at the bottom of the container may move to the top or the part that was at one side may move to the other side. As this mixing takes place, heat transfer occurs by conduction and radiation from one part of the fluid to

another and between the fluid and its surroundings. In other words, convection transports portions of the fluid from one place to another, mixes the fluid, and thus provides an opportunity for heat transfer to occur. But convection does not, in and of itself, "transfer" thermal energy.

Convection serves a vital purpose in bringing the different parts of a fluid into close contact with each other so that heat transfer can occur. Without convection, there would be little heat transfer from, to, or within fluids, since most fluids are very poor at transferring heat except when they are in motion.

Two kinds of convection may be distinguished. *Natural convection* occurs when there are differences in the density of different parts of the fluid. The differences in density are usually caused by unequal temperatures within the mass of fluid. For example; as air passes over a hot radiator, it is heated and becomes less dense and therefore begins to rise. Cooler, heavier air is drawn in to replace the heated air that has moved upward, and convection currents are thus set up. Another example of natural convection, and one that may be quite readily observed, may be found in a pan of water that is being heated on a stove. As the water near the bottom of the pan is heated first, it becomes less dense and moves upward. This displaces the cooler, heavier water and forces it downward; as the cooler water is heated in turn, it rises and displaces the water near the top. By the time the water has almost reached the boiling point, a considerable amount of motion can be observed in the water.

Forced convection occurs when some mechanical device such as a pump or a fan produces movement of a fluid. Many examples of forced convection may be observed in a typical power plant; e.g., feed pumps transporting water to the boilers, circulating water pumps moving cooling water through the main condenser, lubricating oil pumps forcing lubricating oil through coolers, and the exhaust gases of the combustion turbine being forced across the heat transfer tubes inside the HRSG, to name but a few.

5.4 Sensible Heat and Latent Heat

The terms sensible heat and latent heat are often used to indicate the effect that the transfer of heat has upon a substance. The flow of heat from one substance to another is normally reflected in a temperature change in each substance; that is, the hotter substance becomes cooler and the cooler substance becomes hotter. However, the flow of heat is not reflected in a temperature change in a substance which is in the process of changing from one physical state to another. When the flow of heat is reflected in a temperature change, we say that *sensible heat* has been added to or removed from a substance. When the flow of heat is not reflected in a temperature change but is reflected in the changing physical state of a substance, we say that *latent heat* has been added or removed.

Since heat is defined as thermal energy in transition, sensible heat and latent heat must not be regarded as two different kinds of heat. Instead, the terms serve to distinguish between two different kinds of effects produced by the transfer of heat. At a more fundamental level, they indicate something about the manner in which the thermal energy was or will be stored. Sensible heat involves internal kinetic energy and latent heat involves internal potential energy.

The three fundamental physical states of all matter are solid, liquid, and gas (or vapor). The physical state of a substance is closely related to the distance between molecules. The molecules are closest together in solids, farther apart in liquids, and farthest apart in gases. When the flow of heat to a substance is not reflected in a temperature change, we know that the energy is being used to increase the distance between the molecules of the substance and thus change it from a solid to a liquid or from a liquid to a gas. In other words, the addition of heat to a substance that is in process of changing from solid to liquid or from liquid to gas results in an increase in the amount of internal potential energy (latent heat) stored in the substance, but it does not result in an increase in the amount of internal kinetic energy. Only after the change of state has been fully accomplished does the addition of heat result in a change in the amount of internal kinetic energy stored in the substance; hence, there is no temperature change until after the change of state is complete.

In a sense, latent heat may be thought of as the energy price that must be paid for a change of state from solid to liquid or from liquid to gas. But the energy is not lost; rather, it is stored in the substance as internal potential energy. The energy price is 'repaid' when the substance changes back from gas to liquid or from liquid to solid; during these changes of state, the substance gives off heat without any change in temperature.

The amount of latent heat required to cause a change of state, or on the other hand, the amount of latent heat given off during a change of state, varies according to the pressure under which the process takes place. For example, it takes about 970 Btu to change 1 pound of water to steam at atmospheric pressure (14.7 psia); but it takes only 62 Btu to change 1 pound of water to steam at 3200 psia.

Figure 3 illustrates the relationship between sensible heat and latent heat for water at atmospheric pressure. The same type of chart could be developed for other substances, but different amounts of thermal energy would be required for each change of temperature or physical state.

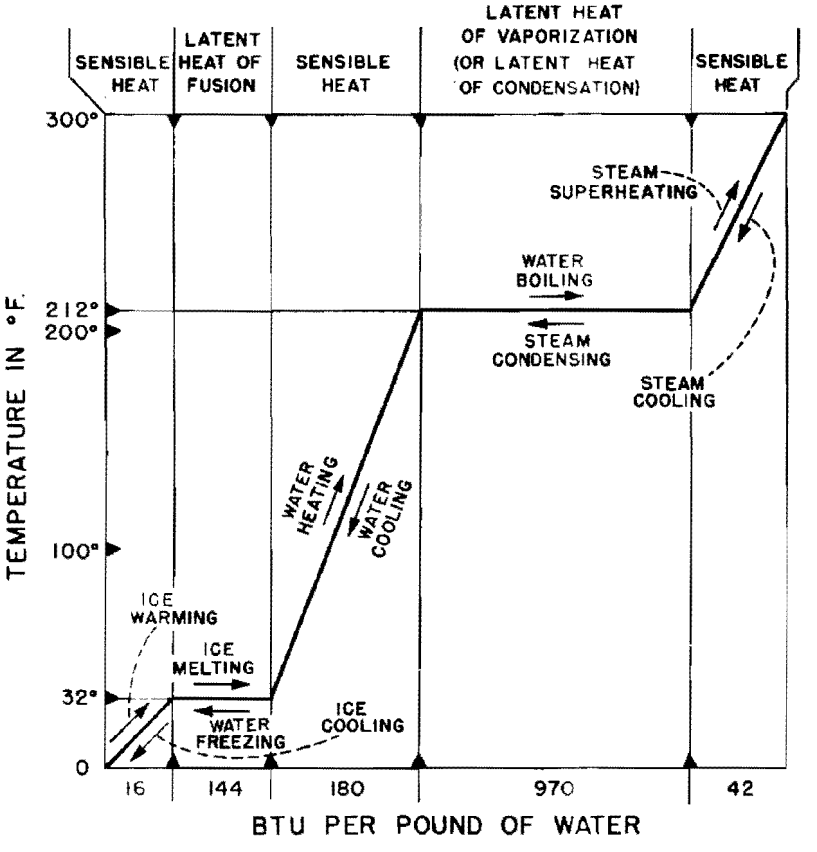

Figure 3 - Relationship Between Sensible Heat and Latent Heat

Starting in the bottom left corner of Figure 3, the process begins with 1 pound of ice at O°F. Moving to the right, the illustration shows that it takes 16 Btu's of energy to raise the temperature of the ice to 32°F. This added energy is referred to as 'sensible heat' since the state of the substance did not change, only its temperature. To change a pound of ice at 32°F to a pound of water at 32°F, 144 additional Btu's of energy must be added. This added energy is referred to as latent heat since it results in a change of state of the substance (from solid to liquid), but not its temperature. After all the ice has melted however, the temperature of the water will be raised as additional heat is supplied. At this point, the added energy is referred to again as sensible heat.

Increasing the temperature of a pound of water from 32°F to a boiling point of 212°F requires 180 Btu's of sensible heat; that is, 1 Btu for each degree of temperature between 32°F and 212°F. To change the state of a pound of water at 212°F to a pound of steam at 212°F, requires an additional 970 Btu's (latent heat). After all the water has been converted to steam, the addition of more heat will cause an increase in the temperature of the steam. For example; adding 42 Btu's to a pound of steam which is at 212°F, increases its superheat temperature to 300°F.

The same relationships apply when heat is being removed. The removal of 42 Btu from a pound of steam which is at 300°F will cause the temperature to drop to 212°F. As the pound of steam at 212°F changes to a pound of water at 212°F, 970 Btu are given off. When a gas or vapor is changing to a liquid, it is usually associated with the term 'latent heat of condensation'. Numerically, of course, the latent heat of condensation is exactly the same as the latent heat of vaporization. The removal of another 180 Btu will lower the temperature of the pound of water from 212°F to 32°F. As the pound of water at 32°F changes to a pound of ice at 32°F, 144 Btu are given off without any accompanying change in temperature. Further removal of heat causes the temperature of the ice to decrease.

5.5 Heat Exchangers

Any device or apparatus designed to allow the flow of thermal energy from one fluid to another is called a heat exchanger. A typical power plant contains an enormous number and variety of heat exchangers, ranging from large items such as boilers and main condensers to relatively small items such as fuel oil heaters and lubricating oil coolers.

As a basis for understanding something about heat transfer in real heat exchangers, it is necessary to visualize the general configuration of the most commonly used type of heat exchanger. With few exceptions, heat exchangers used in a power plant are of the indirect or surface type; that is, heat flows from one fluid to another through some kind of tube, plate, or other surface that separates the two fluids and keeps them from mixing. Most surface heat exchangers are of the shell-and-tube type, consisting of a bundle of metal tubes that fit inside a shell. One fluid flows through the inside of the tubes and the other flows through the shell, around the outside of the tubes.

The exchanges of thermal energy that take place in even a simple heat exchanger are quite complex. The processes of conduction, radiation, and convection are involved in practically all heat exchangers. Processes involving latent heat; that is, the processes of evaporation, condensation, melting, and solidification may contribute to heat transfer problems. In all cases, heat transfer is affected by physical properties of the fluids which are exchanging thermal energy and by physical properties of the metal through which the change is being effected. The physical properties which affect the rate of heat transfer through a heat exchanger include:

- Temperature difference between the two substances
- The extent and nature of the fluid films
- The thickness and nature of the metals through which heat transfer takes place
- The length and area of the path of heat flow
- The types of surfaces involved
- The velocity of flow

Because heat transfer is such a complex phenomenon, heat transfer calculations are necessarily complex. For some purposes, heat transfer problems are simplified by the use of an overall coefficient of heat transfer (U) which may be determined experimentally for any specific set of conditions. Tabulated values of U are available for various kinds of heat exchanger metal tubes, for building materials, and for other materials. In most cases the values of U are approximate, since various conditions such as temperature, velocity of flow, condition of the heat transfer surfaces, and the physical properties of the fluids have a profound effect upon the amount of heat transferred.

The transfer of heat in a heat exchanger involves the flow of heat from the hot fluid to the tube metal, and from the tube metal to the cold fluid. In addition, heat must also be transferred through two layers of fluid (one on the inside and one on the outside of the tube) which are not flowing with the remainder of the fluid but are almost motionless. These relatively stagnant layers, known as 'boundary layers' or 'fluid films' are extremely small in size but have an extremely important effect on heat transfer.

As previously noted, most fluids are very poor conductors of heat. As a fluid is flowing, however, convection and mechanical mixing of the fluid bring the molecules into such intimate contact that heat transfer can and does occur. Other things being equal, increasing the velocity of fluid flow increases heat transfer.

Since the fluid film is almost motionless, heat transfer through the film is very poor. The effect of fluid films on heat transfer is shown in Figure 4. The temperature line indicates the changes in temperature that occur as heat is transferred from the hot fluid to the fluid film, from this fluid film to the tube metal, from the tube metal to the other fluid film, and from this fluid film to the cold fluid. As can be seen, the major part of the temperature drop occurs in the fluid films rather than in the tube metal. Note also, that the thicker fluid film is more resistant to heat transfer than the thinner fluid film.

The velocity of flow and the amount of turbulence in the flow affect heat transfer by altering the thickness of the fluid film. Increasing the velocity of flow diminishes the thickness of the fluid film and thus increases heat transfer. Turbulent flow breaks up the fluid film and thus increases heat transfer. Although there are some obvious disadvantages to excessive turbulence, many heat exchangers are designed to operate with a certain amount of turbulence so that the fluid films are kept to a minimum.

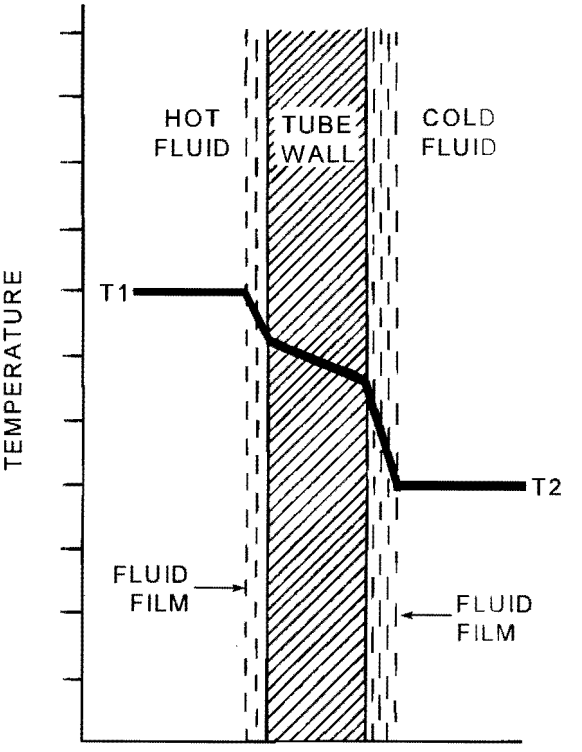

Figure 4 - Heat Transfer Surfaces

In real heat exchangers, the accumulation of deposits of scale, soot, or dirt on the inside or the outside of the tubes has a profound and detrimental effect upon heat transfer. Such deposits not only reduce the efficiency of the heat exchanger but also tend to cause overheating of the tube metal.

In surface heat exchangers, the components may be arranged so as to provide parallel flow, counter flow, or cross flow of the two fluids. Refer to Figure 5. In *parallel flow* both fluids flow in the same direction. Parallel flow heat exchangers are rarely used, largely because they require an impossibly long heat transfer surface to achieve the required amount of heat transfer. In *counter flow* heat exchangers, the two fluids flow in opposite directions. Many heat exchangers used throughout a power plant are of the counter flow type. In *cross flow* heat exchangers, one fluid flows at right angles to the other. Cross flow is used particularly where the purpose of the heat exchanger is to remove latent heat and thus change the physical state of a substance. Main and auxiliary condensers are typically of the cross flow type.

Surface heat exchangers are referred to as single pass units, if each fluid passes the other only once, or as multi-pass units, if one fluid passes the other more than once. Multi-pass flow may be obtained by the arrangement of the tubes and of the fluid inlets and outlets, or it may be obtained by using baffles to guide a fluid so that it passes the other fluid more than once before it leaves the heat exchanger.

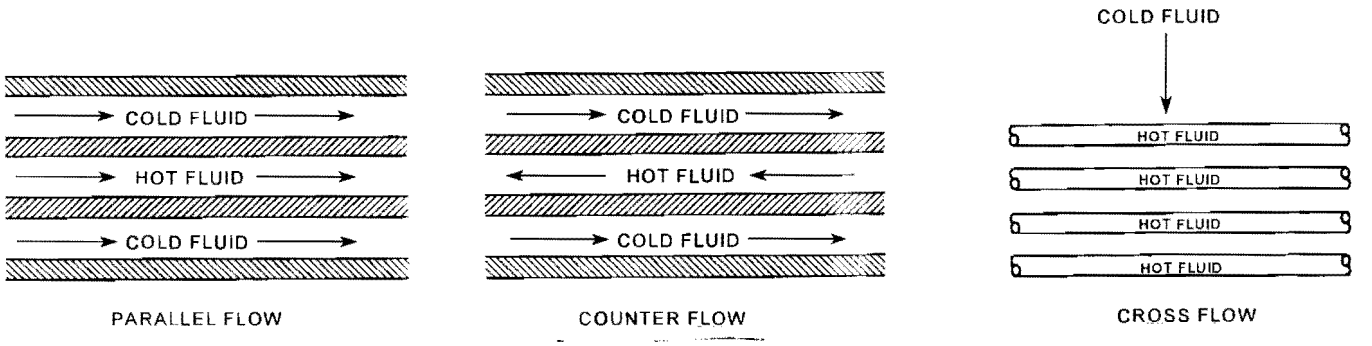

Figure 5 - Heat Exchanger Arrangements

6.0 THERMODYNAMIC PRINCIPLES

6.1 Thermodynamic Systems

A thermodynamic system may be defined as a bounded region which contains matter. The boundaries may be fixed or they may vary in shape, form, and location. The matter within a system may be in any form (solid, liquid, or gas) or in some combination of forms. For some purposes, devices such as engines, pumps, boilers, and so forth may be regarded as being matter included within a thermodynamic system; for other purposes, each such device may be considered as a system in itself. A thermodynamic system may be capable of exchanging energy, in the form of heat and/or work, with its environment; or it may be an isolated system in which case no heat can flow to or from the system and no work can be done on or by the system.

A thermodynamic system requires a working substance to receive, store, transport, and deliver energy. In most systems, the working substance is a liquid, vapor, or gas. The state of a thermodynamic system is specified by giving the values of two or more properties. These properties include such common properties as pressure, temperature, volume, and mass, as well as more complex properties such as enthalpy and entropy (discussed later). Although some systems are adequately described by giving the value of only two variables, many systems require the specification of three or more variables.

6.2 Thermodynamic Processes

A thermodynamic process may be defined as; *any physical occurrence during which an effect is produced by the transformation or redistribution of energy*. The occurrence of a thermodynamic process is evidenced by changes in some or all of the state variables of the system. The processes of most interest in power plant engineering are those involving heat and work.

In connection with any process, it is usually necessary to consider the physical character of the process. To include; the manner in which energy is transformed or redistributed as the process takes place; the type and amount of energy that is stored in the system before and after the process; and the changes which are brought about in the system as a result of the process. It is also necessary to consider the energy exchanges that occur between the system and its surroundings during the process, since such energy exchanges will have an effect on the final state of the system.

The lifting of an object; for example, the lifting of a rock from the base of a cliff to the top of the cliff; is a simple example of a process involving work against gravity. Before the process begins, the energy required to lift the rock is stored in some form in some other energy system. While the process is occurring, energy in the form of work flows from the external system to the earth-rock system. At the end of the

process, the energy is stored in the earth-rock system in the form of mechanical potential energy. The change which has been brought about by this process is manifested by the separation of the rock and the earth.

Next, consider that the rock is pushed off the top of the cliff and allowed to fall freely toward the base of the cliff. Disregarding the push (which is actually an input of energy from some external system), the process which now takes place is an example of work done by gravity. The work done by gravity converts the mechanical potential energy of the system into mechanical kinetic energy. Thus it is clear that energy in transition, work in this case, begins and ends as stored energy.

When the rock hits the earth, other processes occur. Some work will be expended in compressing the earth upon which the rock falls, and some energy will then be stored as internal kinetic energy in the rock and in the earth. The increase in internal kinetic energy will be manifested by a rise in the temperature of the rock and of the earth, and still another process will then take place as heat flows from the rock and from the earth. Some energy may also be stored as internal potential energy because of molecular displacements in the rock and the earth.

Another type of thermodynamic process is elastic deformation; for example, the compression of a spring. As force is applied to compress the spring, work is done. The major effect of the energy supplied as work is to decrease the distance between molecules in the spring, thus increasing the amount of internal potential energy stored in the spring. If the spring is suddenly released, the stored internal potential energy is suddenly released as is evident by the rapid uncoiling of the spring.

A third example of a thermodynamic process is one of work against friction, which can be illustrated by dragging a solid body across a rough horizontal surface. The work done in moving the object will be equal to the force required to overcome the friction multiplied by the distance through which the object is moved. In this process, the energy supplied as work is transformed very largely into internal kinetic energy, as evidenced by an increase in temperature. Some of the energy may be transformed into internal potential energy because of molecular displacements in the object and in the surface over which it is being moved.

Finally, the addition of thermal energy to a piece of metal is a simple example of a thermodynamic process involving heat. As the metal is heated, the temperature rises, indicating a storage within the metal of internal kinetic energy. Also, the metal expands; thus part of the energy delivered as heat is transformed into work as the metal expands against the resistance of its surroundings. When heating the metal to its melting point, a process occurs in which the flow of heat results in a change in the physical state of the substance but does not, at this point, result in a further rise in temperature.

6.3 The Principle of Energy Conservation

Throughout previous discussions on energy, many concepts and general principles have been used in describing terms such as p-v energy, internal energy, work, etc. One of the principles which has been used extensively is the principle of energy conservation. This principle may be stated in several ways, each arriving at ultimately the same meaning and conclusion. Most commonly, the principle of energy conservation may be stated as:

- Energy can not be created or destroyed, only transformed.
- Energy may be transformed from one form to another, but the total energy of any body or system of bodies is a quantity that can neither be increased nor diminished by the action of the body or bodies.
- The total quantity of energy in the universe is always the same.

Regardless of the mode of expression, the principle of energy conservation applies to all kinds of energy. Energy equations for many thermodynamic processes are based directly upon the principle of energy conservation. When the principle of energy conservation is written in equation form, it is known as 'the general energy equation' and is expressed as:

Energy in = Energy out

In more detail, the general energy equation may be stated as; the energy entering a system equals the energy leaving the system plus any accumulation and minus any decline in the amount of energy stored within the system.

6.4 The First Law Of Thermodynamics

The first law of thermodynamics deals with the transformation of mechanical energy to thermal energy, and of thermal energy to mechanical energy. The first law of thermodynamics is commonly stated as follows; *thermal energy and mechanical energy are mutually convertible, in the ratio of 778 foot-pounds to 1 Btu.*

The ratio of conversion between mechanical energy and thermal energy is known as the mechanical equivalent of heat. The mechanical equivalent of heat provides us, directly or by extension, with a number of useful numerical values relating to heat, work, and power. Some of the most widely used values are given below; others may be obtained from engineering handbooks and similar publications.

1 Btu	= 778 ft-lb
1 hp	= 33,000 ft-lb per min *or* 550 ft-lb per sec
1 kw	= 1.341 hp
1 hp	= 2545 Btu per hr *or* 42.42 Btu per min
1 kw	= 3413 Btu per hr
1 kw	= 44,256 ft-lb per min
1 kw/hr	= 3413 Btu

Another common statement of the first law of thermodynamics is that a perpetual motion machine of the first class is impossible. To understand the significance of this statement, it is necessary to understand the classification of perpetual motion machines. Although no perpetual motion machine exists, or indeed has ever been constructed, it is possible to conceive of three different categories; each described below.

A perpetual motion machine of the first class is one which would put out more energy in the form of work than it absorbed in the form of heat. Since such a machine would actually create energy, it would violate the first law of thermodynamics and the principle of the conservation of energy. A perpetual motion machine of the second class would permit the reversal of irreversible processes and would thus violate the second law of thermodynamics, as discussed presently. A machine of the third class would be one in which absolutely no friction existed. Interestingly enough, there are no theoretical grounds for declaring that a machine of the third class is completely impossible; however, such a machine would be entirely contrary to our experience and would violate some of our profoundest convictions about the nature of energy and matter.

6.5 Thermodynamic Cycles

A thermodynamic cycle is a recurring series of thermodynamic processes through which an effect is produced by the transformation or redistribution of energy. In other words, a cycle is a series of processes repeated over and over again in the same order.

All thermodynamic cycles may be classified as being open cycles or closed cycles. An open cycle is one in which the working fluid is taken in, used, and then discarded. A closed cycle is one in which the

working fluid never leaves the cycle, except through accidental leakage; instead, the working fluid undergoes a series of processes which are of such a nature that the fluid is returned periodically to its initial state and is then used again.

An example of an *open cycle* can be represented by an internal combustion engine, in which atmospheric air supplies the oxygen for combustion and in which the exhaust products are returned to the atmosphere. In simple terms, an open cycle can be defined as a cycle which includes the atmosphere at some point.

An example of a *closed cycle* can be represented by the main steam system of a typical power plant. In such a cycle, the working substance (water) is changed to steam in the boilers. The steam performs work as it expands through the turbines, and is then condensed to water again in the condenser. The water is returned to the boilers as boiler feed, and is thus used over and over again.

Thermodynamic cycles are also classified as heated-engine cycles or as unheated-engine cycles, depending upon the point in the cycle at which heat is added to the working substance. In a *heated-engine cycle*, heat is added to the working substance in the engine itself. An internal combustion engine has a heated-engine cycle. In an *unheated-engine cycle*, the working substance receives heat in some device which is separate from the engine. The condensing steam power plant has an unheated-engine cycle, since the working substance is heated separately in the boilers and then piped to the engines (steam turbines).

There are five basic elements in any thermodynamic cycle; (1) the working substance, (2) the engine, (3) a heat source, or high temperature region, (4) a heat receiver, or low-temperature region, and (5) a pump.

The working substance is the medium by which energy is carried through the cycle. The engine is the device which converts the thermal energy of the working substance into useful mechanical energy in the form of work. The heat source supplies heat to the working substance. The heat receiver absorbs heat from the working substance. The pump moves the working substance from the low pressure side of the cycle to the high pressure side.

The essential elements of a closed, unheated-engine cycle are shown in Figure 6. This type of thermodynamic cycle is the basic arrangement of a typical condensing steam power plant.

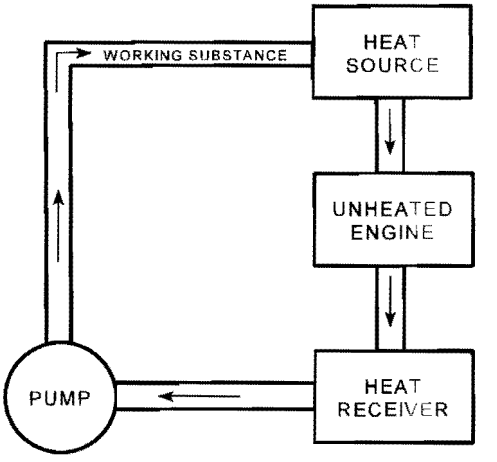

Figure 6 - Closed Unheated-Engine Cycle

6.6 The Concept of Reversibility

When a pan of water is placed on a stove and heat is applied, it is expected that the water will boil rather than to freeze. When hot and cold water are added to a single container, it is not expected that the resulting mixture will resolve itself into two separate batches of water at two different temperatures. When a valve is opened on a cylinder of compressed air, it is expected that compressed air will rush out; not rush in. When a block of wood is dragged across a rough surface, it is expected that some of the mechanical energy expended in this act to be converted into thermal energy; that is, we expect a storage of internal energy in the wooden block and the rough surface, as evidenced by temperature rises in these materials. It would seem rather impossible then, if this stored internal energy should suddenly change form and move the wooden block back to its original position(?).

Each of the above scenario's illustrate that we have certain expectations, based on experience, as to the direction in which processes will move. The reasoning of our expectations is attested by the fact that in

all recorded history there is no report of water freezing instead of boiling when heat is applied; there is no report of a lukewarm fluid unmixing itself and separating into hot and cold fluids; there is no report of a gas compressing itself without the agency of some external force; there is no report of the heat of friction being spontaneously utilized to perform mechanical work.

Are these actions really impossible? The first law of thermodynamics says that mechanical energy and thermal energy are mutually convertible, but it says nothing about the direction of such conversions. If we consider only the first law, all the improbable actions just mentioned are perfectly possible and all processes could be thought of as being reversible. In an absolute sense, perhaps, we cannot guarantee that water will never freeze instead of boil when it is placed on a hot stove; but we are certainly safe in saying that this or any other completely reversible thermodynamic process is at the outer limits of probability. For all practical purposes then, it is true that there is no such thing as a completely reversible process.

Nevertheless; the concept of reversibility is extremely useful in evaluating real thermodynamic processes. With this information in mind, a reversible thermodynamic process can be defined as one which would have the following characteristics; (1) the process could be made to occur in precisely reverse order, so that the energy system and all associated systems would be returned from their final condition to the conditions that existed before the process started; and (2) all energy that was transformed or redistributed during the process would be returned from its final to its original form, amount, and location.

6.7 The Second Law Of Thermodynamics

Since the first law of thermodynamics does not deal with the direction of thermodynamic processes, and since experience indicates that actual processes are not reversible, it is apparent that the first law must be supplemented by some statement of principle that will limit the direction of thermodynamic processes. The second law of thermodynamics is such a statement. Although the second law is perhaps more empirical than the first law, and perhaps something less of a "law" in an absolute sense, it is of enormous practical value in the study of thermodynamics.

The second law of thermodynamics may be stated in various ways. One statement, known as the Clausius statement, is that *no process is possible where the sole result is the removal of heat from a low temperature reservoir and the absorption of an equal amount of heat by a high temperature reservoir*. Among other things, this statement indicates that water will not freeze when heat is applied. Note that the Clausius statement includes and goes somewhat beyond the common observation that heat flows only from a hotter to a colder substance.

The Kelvin-Planck statement is another statement of the second law of thermodynamics; *no process is possible where the sole result is the removal of heat from a single reservoir and the performance of an equivalent amount of work*. Among other things, this statement says that we cannot expect the heat of friction to reverse itself and perform mechanical work. More broadly, energy in the form of work can be converted entirely to energy in the form of heat; but energy in the form of heat can never be entirely converted to energy in the form of work.

A very important deduction to be drawn from the second law is that; *no engine, actual or ideal, can convert all the heat supplied to it into work, since some heat must always be rejected to a receiver which is at a lower temperature than the source*. In other words, there can be no heat flow without a temperature difference and there can be no conversion to work without a flow of heat. A further deduction from this is sometimes given as a statement of the second law; no thermodynamic cycle can have a thermal efficiency of 100 percent.

In conclusion, it can be said that the first law of thermodynamics deals with the conservation of energy and with the mutual convertibility of heat and work, while the second law limits the direction of thermodynamic processes and the extent of heat-to-work energy conversions.

6.8 The Concept of Entropy

Entropy is a measure of the unavailable energy contained within a working fluid or a thermodynamic system. Entropy is known to exists because when experiments are conducted in laboratories, some of the energy that is know to be present cannot be measured. In this respect, entropy is considered to be a property or characteristic of a substance that drives the heat transfer process.

The concept of reversibility and the second law of thermodynamics are closely related to the concept of entropy. In fact, the second law may be stated as; *no process can occur in which the total entropy of an isolated system decreases; the total entropy of an isolated system can theoretically remain constant in some reversible (ideal) processes, but in all irreversible (real processes) the total entropy of an isolated system must increase.*

From other statements of the second law, it is known that the transformation of heat to work is always dependent upon a flow of heat from a high temperature region to a low temperature region. The concept of the unavailability of a certain portion of the energy supplied as heat to any thermodynamic system is clearly implied in the second law, since it is apparent that some heat must always be rejected to a receiver which is at a lower temperature than the source, if there is to be any conversion of heat to work. The heat which must be so rejected is therefore unavailable for conversion into mechanical work.

Entropy is an index of the unavailability of energy. Since heat can never be completely converted into work, entropy may be thought of as a measure or an indication of how much heat must be rejected to a low temperature receiver if we are to utilize the rest of the heat for the production of useful work. We may also think of entropy as an index or measure of the reversibility of a process. All real processes are irreversible to some degree, and all real processes involve a growth or increase of entropy. Irreversibility and entropy are closely related; any process in which entropy has increased is an irreversible process.

Entropy changes depend upon the amount of heat transferred to or from the working fluid, upon the absolute temperature of the heat source, and upon the absolute temperature of the heat receiver. The fact that the total entropy of an isolated system must always increase does not mean that the entropy of all parts of the system must always increase. In many real processes, we find increases in entropy in some parts of a system and, at the same time, decreases in entropy in other parts of the system. But the important thing to note is that the increases in entropy are always greater than the decreases; therefore, the total entropy of an isolated system must always increase.

Each increase in entropy is permanent. In a universal sense, entropy can be created but it can never be destroyed or gotten rid of, although it may be transferred from one system to another. Every natural process that occurs in the universe increases the total entropy of the universe, and this increase in entropy is irreversible. The concept of the universe eventually "running down" might be expressed in terms of entropy by saying that the entropy of the universe is constantly "building up". The so called "heat death of the universe" is envisioned as the ultimate result of all possible natural processes having taken place and the universe being in total equilibrium, with entropy at the absolute maximum. Such a statement need not imply a total lack of energy remaining in the universe; but any energy that might remain would be completely unavailable and therefore completely useless.

7.0 WORKING SUBSTANCES

As previously discussed, a thermodynamic system requires a working substance to receive, store, transport, and deliver energy. The working substance is almost always a fluid and is therefore frequently referred to as the *working fluid*. Water (together with its vapor, steam) is one of the most commonly used working fluids; although air, ammonia, carbon dioxide, and a wide variety of other fluids are used in certain kinds of systems. A working substance may change its physical state during the course of a thermodynamic cycle or it may remain in one state, depending upon the nature of the cycle and the processes involved.

7.1 Liquids and Their Vapors

When heat is transferred to a liquid, the average velocity of the molecules is increased and the amount of internal kinetic energy stored in the liquid is increased. As the average velocity of the molecules increases, some molecules which are at or near the surface of the liquid momentarily achieve unusually high velocities; and some of these escape from the liquid and enter the space above, where they exist in the vapor state. As more and more of the molecules escape and come into the vapor state, the probability increases that some of the vapor molecules will momentarily have unusually low velocities; these molecules will be captured by the liquid. As a result of this exchange of molecules between the liquid and the vapor, a condition of equilibrium is reached and an equilibrium pressure is established. The equilibrium pressure depends upon the molecular structure of the fluid and upon its temperature. Therefore; for any given fluid there is a definite relationship between the temperature and the pressure at which a liquid and its vapor may exist in equilibrium contact with each other.

As long as the vapor is in contact with the liquid from which it is being generated, the liquid and the vapor will remain at the same temperature. If the liquid and the vapor are in a closed container (such as a boiler with all steam stop valves closed) both the temperature and the pressure of the liquid and its vapor will increase as heat is added. If the vapor is permitted to leave the steam space at a rate equal to the evaporation rate, an equilibrium will be established at the equilibrium pressure for the particular temperature.

The pressure and the temperature which are related in the manner just described are known as the *saturation pressure* and the *saturation temperature*. Thus, for any specified pressure there is a corresponding temperature of vaporization known as the saturation temperature; and for any specified temperature there is a corresponding saturation pressure.

A vapor which is under any specified pressure and at the saturation temperature corresponding to that pressure is said to be a *saturated vapor*. Thus, water at 14.7 psia and 212°F produces a vapor known as *saturated steam*. As previously noted, it is impossible to raise the temperature of a vapor above the temperature of its liquid as long as the two are in contact. If the vapor is drawn off into a separate container, however, and additional heat is supplied to the vapor, the temperature of the vapor is raised. A vapor which has been raised to a temperature that is above its saturation temperature is called a *superheated vapor*. The vessel or container in which the saturated steam is superheated is called a superheater.

The elementary boiler and superheater illustrated in Figure 7 shows the general principle of generating and superheating steam. Practically all power plant boilers have superheaters for superheating the saturated steam generated in the generating sections of the boiler. The amount by which the temperature of a superheated vapor exceeds the temperature of a saturated vapor at the same pressure is known as the *degree of superheat*. For example, if saturated steam at a pressure of 600 psia and a corresponding saturation temperature of 486°F is superheated to 786°F, the degree of superheat is 300°F.

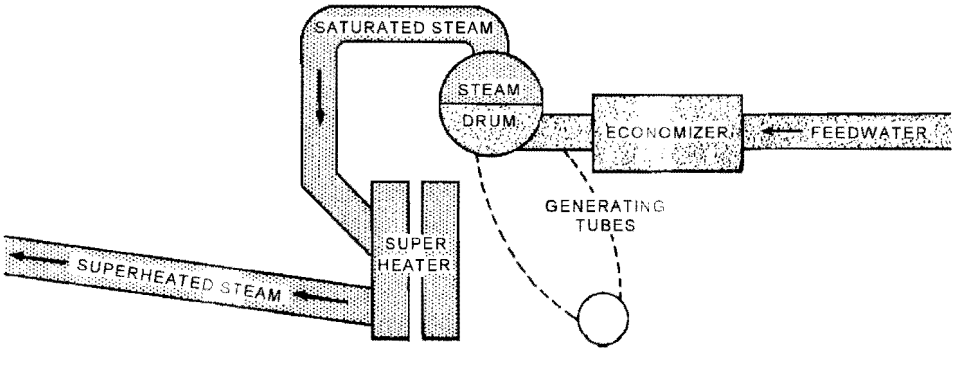

Figure 7 - Elementary Boiler And Superheater

7.2 Critical Point

For any substance there is a *critical point* at which the properties of the saturated liquid are exactly the same as the properties of the saturated vapor. For water, the critical point is reached at 3206.2 psia (critical pressure) and 704.4°F (critical temperature). At the critical point, the vapor and the liquid are indistinguishable. No change of physical state occurs when the pressure is increased or when additional heat is supplied; the vapor cannot be made to liquefy and the liquid cannot be made to vaporize as long as the substance is at or above its critical pressure and critical temperature. At this point, it can no longer be referred to as 'water' and/or 'steam' since it is not possible to tell the water and the steam apart. Instead, the substance is merely called a 'fluid' or a 'working fluid'.

7.3 Enthalpy

The condition of a working fluid at any point within a thermodynamic cycle is established by the properties of the substance at that point. The properties that are of special interest in power plant thermodynamics include pressure, temperature, volume, internal energy, entropy, and enthalpy.

In previous discussions, *internal energy* was defined as the total sum of the molecular movement within a substance due to the presence of heat. As more and more heat is added to a substance, the molecular movement within the substance increases; and the internal energy of the substance increases proportionately. *Pressure-volume energy* is represented by the expansion that occurs within a substance when it is heated. In the analysis of energy transfer systems, the internal energy and the pressure-volume energy often occur together in expressions describing the energy possessed by the working fluid. *The enthalpy of a substance is the sum of its internal energy and its pressure-volume energy*. In general terms, enthalpy is a measure of how much energy a substance contains that can be used to perform work.

Enthalpy is expressed by the equation: $h = u + (Pv \div 778)$

where;
- h = enthalpy
- u = internal energy
- Pv = pressure-volume energy (pressure x specific volume)
- 778 = conversion factor for foot-pounds per Btu

Enthalpy is a property of a substance like pressure, temperature, and volume; but it cannot be measured directly. Normally, the enthalpy of a substance is given with respect to some reference value. For example, the specific enthalpy of water or steam is given using the reference that the specific enthalpy of water is zero at 32°F @ atmospheric pressure. The fact that the absolute value of specific enthalpy is unknown is not a problem; however, since it is the change in specific enthalpy and not the absolute value that is important in practical problems.

One way to quantify the definition of enthalpy is to consider the change in enthalpy of water when it is heated at standard atmospheric pressure. When energy is added to a system of water in the form of heat, the amount of heat added can be quantified in terms of the change in temperature of the water. In doing so, there is no change in the potential energy or the kinetic energy of the water when the heat is added. In addition, the pressure remains constant and there is no appreciable change in volume; thus, there is no appreciable change in P-V energy of the water. Therefore; the energy added in the form of heat must be reflected as a change in the internal energy of the fluid. Thus; for this case, the heat added equals the change in internal energy. But since the change in P-V energy is essentially zero, the change in internal energy equals the change in enthalpy; and the heat added equals the change in enthalpy. Thus, the change in enthalpy equals the heat added to the system, which equals the product of the mass, the specific heat and the change in temperature.

Different substances have different amounts of internal energy for the same temperature. For example, to increase the temperature of 1 pound of water at a given temperature by 1 degree Fahrenheit, it takes 1 British thermal unit (Btu). However, to increase the temperature of 1 pound of steel at the same temperature by 1°F, it takes 0.118 Btu; 1 pound of petroleum takes 0.5 Btu. Therefore, 1 pound of water has more internal energy than 1 pound of steel or petroleum at the same temperature. Water is a liquid, and a liquid is at a higher internal energy level than a solid such as steel. Also, a gas is at a higher internal energy level than a liquid. The molecules of the gas are in motion and freer to move than those of a liquid. The molecules of a liquid are freer to move than those of a solid. Energy must be added to convert a liquid such as water to a gas such as steam, or to convert a solid such as ice to water. As the energy is added and the substance goes from solid to liquid and then from liquid to gas, the molecular structure becomes more random. The molecules must find room to move.

Because the gas molecules are at a higher energy level and are free to separate and move, gases like air or steam are compressible. This means their volume can be greatly reduced if put under pressure, and this pressure can be put to work through expansion. Gases under pressure can expand greatly to do work. Since gases are compressible and can therefore retain pressure-volume energy, they also have a greater *enthalpy* than a solid or liquid. Solids and liquids are normally considered incompressible, so their pressure-volume energy is zero.

7.4 Steam Tables

It is well known that water boils and forms steam at a temperature of 212°F (100°C) when the water and steam are at normal atmospheric pressure (14.7 psia). If water is heated under pressure, the temperature at which it boils increases. If the pressure on the water is reduced, the temperature at which it boils is reduced. For each particular pressure, there is therefore a definite saturation temperature at which, if heat is added, the water boils and forms steam; or, if heat is removed, the steam condenses to form water.

Because of the complicated equations needed to describe the properties of vapors with respect to thermal energy, engineers customarily depend upon tables of vapor properties for information concerning the properties of liquids and their vapors. The vapor tables that are perhaps most commonly used are those which give the thermodynamic properties of steam.

Steam tables list the changes of each property water/steam with the resultant changes of others. Because the properties of water/steam are dependant on each other, a change in one property results in a change of the others. The two steam tables most commonly used in power plant engineering are the 'saturated steam table' and the 'superheated steam table'. Figure 8 is an excerpt from a saturated steam table. Saturated steam tables give the energy transfer properties of saturated water/steam for temperatures from 32°F to 705.47°F (the critical point) and for the corresponding pressures from 0.08849 to 3208.2 psi.

	Abs Press	Specific Volume			Enthalpy			Entropy		
Temp Fahr t	Lb per Sq In p	Sat. Liquid v_f	Evap v_{fg}	Sat Vapor v_g	Sat. Liquid h_f	Evap h_{fg}	Sat Vapor h_g	Sat Liquid s_f	Evap s_{fg}	Sat Vapor s_g
460.0	466.87	0.01961	0.97463	0.99424	441.5	763.2	1204.8	0.6405	0.8299	1.4704
464.0	485.56	0.01969	0.93588	0.95557	446.1	758.6	1204.7	0.6454	0.8213	1.4667
468.0	504.83	0.01976	0.89885	0.91862	450.7	754.0	1204.6	0.6502	0.8127	1.4629
472.0	524.67	0.01984	0.86345	0.88329	455.2	749.3	1204.5	0.6551	0.8042	1.4592
476.0	545.11	0.01992	0.82958	0.84950	459.9	744.5	1204.3	0.6599	0.7956	1.4555
480.0	566.15	0.02000	0.79716	0.81717	464.5	739.6	1204.1	0.6648	0.7871	1.4518
484.0	587.81	0.02009	0.76613	0.78622	469.1	734.7	1203.8	0.6696	0.7785	1.4481
488.0	610.10	0.02017	0.73641	0.75658	473.8	729.7	1203.5	0.6745	0.7700	1.4444
492.0	633.03	0.02026	0.70794	0.72820	478.5	724.6	1203.1	0.6793	0.7614	1.4407
496.0	656.61	0.02034	0.68065	0.70100	483.2	719.5	1202.7	0.6842	0.7528	1.4370
500.0	680.86	0.02043	0.65448	0.67492	487.9	714.3	1202.2	0.6890	0.7443	1.4333
504.0	705.78	0.02053	0.62938	0.64991	492.7	709.0	1201.7	0.6939	0.7357	1.4296
508.0	731.40	0.02062	0.60530	0.62592	497.5	703.7	1201.1	0.6987	0.7271	1.4258
512.0	757.72	0.02072	0.58218	0.60289	502.3	698.2	1200.5	0.7036	0.7185	1.4221
516.0	784.76	0.02081	0.55997	0.58079	507.1	692.7	1199.8	0.7085	0.7099	1.4183

Figure 8 - Saturated Steam Table

The information given in each column of the steam table (Figure 8) for the properties of saturated steam is described briefly below. Note that the subscript *f* is commonly used to denote properties of the saturated liquid, *g* to denote properties of the saturated vapor, and *fg* to denote property changes between the two states.

Temperature (*t*) - Lists the saturation temperature (°F) for that line item in the steam table. This temperature is commonly referred to as 'the boiling point' of the liquid at the pressure shown in the next column (absolute pressure). The temperature is usually given in degrees Fahrenheit rather than in degrees of an absolute temperature scale. However, the absolute temperature can always be obtained by simple computation if it should be needed.

Absolute Pressure (*p*) - Lists the saturation pressure of the saturated water/steam at the given temperature. Note that the pressure is absolute pressure, not gauge pressure.

Specific Volume - Lists the specific volume (cubic feet per pound mass) of the working fluid at three different states; saturated liquid (*Vf*), evaporation (*Vfg*), and saturated vapor (*Vg*). These figures represent the expansion of the working fluid as it changes from a liquid to a vapor.

Enthalpy - Lists the enthalpy (Btu per pound mass) of the working fluid at three different states:

- Saturated Liquid (*Hf*) - Gives the enthalpy per pound of the saturated liquid at the pressure and temperature shown. The enthalpy of saturated water at 32°F and the corresponding saturation pressure of 0.08854 psia is taken as zero; hence, all enthalpy figures indicate enthalpy with respect to this arbitrarily assigned zero point. For example, the enthalpy of 1 pound of saturated water at 190 psia and 377.51°F is 350.79 Btu more than the enthalpy of 1 pound of saturated water at 0. 08854 psia and 32°F.
- Evaporation (*Hfg*) - Gives the enthalpy of evaporation per pound of working fluid; that is, the change in enthalpy that occurs during evaporation. This column is of particular significance since it indicates the Btu per pound that must be supplied to change the saturated liquid (water) to the saturated vapor (steam) at the pressure and temperature shown. In other words, the enthalpy of evaporation was formerly described as the latent heat of vaporization.
- Saturated Vapor (*Hg*) - Gives the enthalpy per pound of the saturated vapor at the pressure and temperature shown. Note that this is the sum of the enthalpy of the saturated liquid and the enthalpy of evaporation.

Entropy - Lists the entropy (Btu per pound mass) of the working fluid at three different states:

- Saturated Liquid (*Hf*) - Gives the entropy per pound of the saturated liquid at the pressure and temperature shown. The zero point for entropy, like the zero point for enthalpy, is arbitrarily established at 32°F and the corresponding saturation pressure of 0.08854 psia.
- Evaporation (S*fg*) - Gives the entropy of evaporation per pound of working fluid at the pressure and temperature shown. In other words, this column shows the change of entropy that occurs during evaporation.
- Saturated Vapor (S*g*) - Gives the entropy per pound of the saturated vapor at the pressure and temperature shown. Note that this is the sum of the entropy of the saturated liquid and the entropy of evaporation.

Tables are also available for the different properties of superheated steam. Superheated steam tables give the energy transfer properties of superheated steam as a function of pressure and temperature. Figure 9 is an excerpt of a typical superheated steam table. For each value of pressure, the saturation temperature is given as well as the values of specific volume, specific enthalpy, and specific entropy at the saturation temperture for both saturated water and saturated steam and at temperatures above saturation for superheated steam. The number of degress of superheat, represented by Sh, is given at each temperature and pressure for superheated steam.

Abs Press. Lb/Sq In. (Sat. Temp)		Sat. Water	Sat. Steam	Temperature – Degrees Fahrenheit 450	500	550	600	650	700	800	900	1000	1100	1200	1300	1400	1500
	Sh			5.40	55.40	105.40	155.40	205.40	255.40	355.40	455.40	555.40	655.40	755.40	855.40	955.40	1055.40
400	v	0.01934	1.1610	1.1738	1.2841	1.3836	1.4763	1.5646	1.6499	1.8151	1.9759	2.1339	2.2901	2.4450	2.5987	2.7515	2.9037
(444.60)	h	424.17	1204.6	1208.8	1245.1	1277.5	1307.4	1335.9	1363.4	1417.0	1470.1	1523.3	1576.9	1631.2	1686.2	1741.9	1798.2
	s	0.6217	1.4847	1.4894	1.5282	1.5611	1.5901	1.6163	1.6406	1.6850	1.7255	1.7632	1.7988	1.8325	1.8647	1.8955	1.9250
	Sh			60	50.60	100.60	150.60	200.60	250.60	350.60	450.60	550.60	650.60	750.60	850.60	950.60	1050.60
420	v	0.01942	1.1057	1.1071	1.2148	1.3113	1.4007	1.4856	1.5676	1.7258	1.8795	2.0304	2.1795	2.3273	2.4739	2.6196	2.7647
(449.40)	h	429.56	1204.7	1205.2	1242.4	1275.4	1305.8	1334.5	1362.3	1416.2	1469.4	1522.7	1576.4	1630.8	1685.8	1741.6	1798.0
	s	0.6276	1.4802	1.4808	1.5206	1.5542	1.5835	1.6100	1.6345	1.6791	1.7197	1.7575	1.7932	1.8269	1.8591	1.8899	1.9195
	Sh				45.97	95.97	145.97	195.97	245.97	345.97	445.97	545.97	645.97	745.97	845.97	945.97	1045.97
440	v	0.01950	1.0554		1.1517	1.2454	1.3319	1.4138	1.4926	1.6445	1.7918	1.9363	2.0790	2.2203	2.3605	2.4998	2.6384
(454.03)	h	434.77	1204.8		1239.7	1273.4	1304.2	1333.2	1361.1	1415.3	1468.7	1522.1	1575.9	1630.4	1685.5	1741.2	1797.7
	s	0.6332	1.4759		1.5132	1.5474	1.5772	1.6040	1.6286	1.6734	1.7142	1.7521	1.7878	1.8216	1.8538	1.8847	1.9143
	Sh				41.50	91.50	141.50	191.50	241.50	341.50	441.50	541.50	641.50	741.50	841.50	941.50	1041.50
460	v	0.01959	1.0092		1.0939	1.1852	1.2691	1.3482	1.4242	1.5703	1.7117	1.8504	1.9872	2.1226	2.2569	2.3903	2.5230
(458.50)	h	439.83	1204.8		1236.9	1271.3	1302.5	1331.8	1360.0	1414.4	1468.0	1521.5	1575.4	1629.9	1685.1	1740.9	1797.4
	s	0.6387	1.4718		1.5060	1.5409	1.5711	1.5982	1.6230	1.6680	1.7089	1.7469	1.7826	1.8165	1.8488	1.8797	1.9093

Figure 9 - Superheated Steam Table

8.0 ENERGY RELATIONSHIPS IN THE POWER PLANT CYCLE

At the beginning of this module, it was stated that every power plant may be thought of as a series of devices and arrangements for the exchange and transformation of energy. Many of these transformations and exchanges have been discussed, but they have not been taken up in sequence. Figure 10 illustrates the basic energy cycle of a conventional power plant utilizing a boiler and a steam turbine generator.

8.1 Power Plant Energy Cycle

The first energy transformation occurs when fuel is burned in the boiler furnace, or in case of an HRSG, from that of the gas turbine exhaust. By the process of combustion, the chemical energy stored in the fuel is transformed into thermal energy. Thermal energy flows from the hot combustion gases to the water in the boiler. While the boiler stop valves are still closed, steam begins to form in the boiler; the volume of the steam remains constant but the pressure and temperature increase, indicating a storage of internal energy. When operating pressure is reached and the steam stop valves are opened, the high pressure of the steam causes it to flow to the turbines toward an area of lower pressure.

At the steam turbine, the pressure of the steam provides the potential for doing work; i.e., the actual conversion of heat to work takes place in the turbines. The changes in internal energy between the boiler and the condenser (as evidenced by changes in pressure and temperature) indicate that heat has been converted to work in the turbines. The work output of the turbines turns the generator shaft to produce electricity.

Two main energy transformations are involved in converting thermal energy to work in the turbines. First, the thermal energy of the steam is transformed into mechanical kinetic energy as the steam flows through the successive stages of nozzles. Second, the mechanical kinetic energy of the steam is transformed into work as the steam impinges upon the projecting blades of the turbine and thus causes the turbine to rotate. The rotation of the turbine rotor provides for the tansformation of mechanical work into electrical energy with the generator. Once through the last stage of the turbine, the steam exhausts into the condenser where it gives up its latent heat of condensation to the circulating water.

For the remainder of the cycle, energy is required to get the water (condensate and feed water) back to the boiler where it will again be heated and changed into steam. Pumps are used to perform this function and they may be either steam or motor driven. In the case of turbine driven feed pumps, the conversion of thermal energy to mechanical energy occurs in the same way as it does in any steam turbine. In the case of motor driven pumps, the energy conversion is from thermal energy to electrical energy (in a turbine generator) and then from electrical energy to mechanical energy (work) in the pumps.

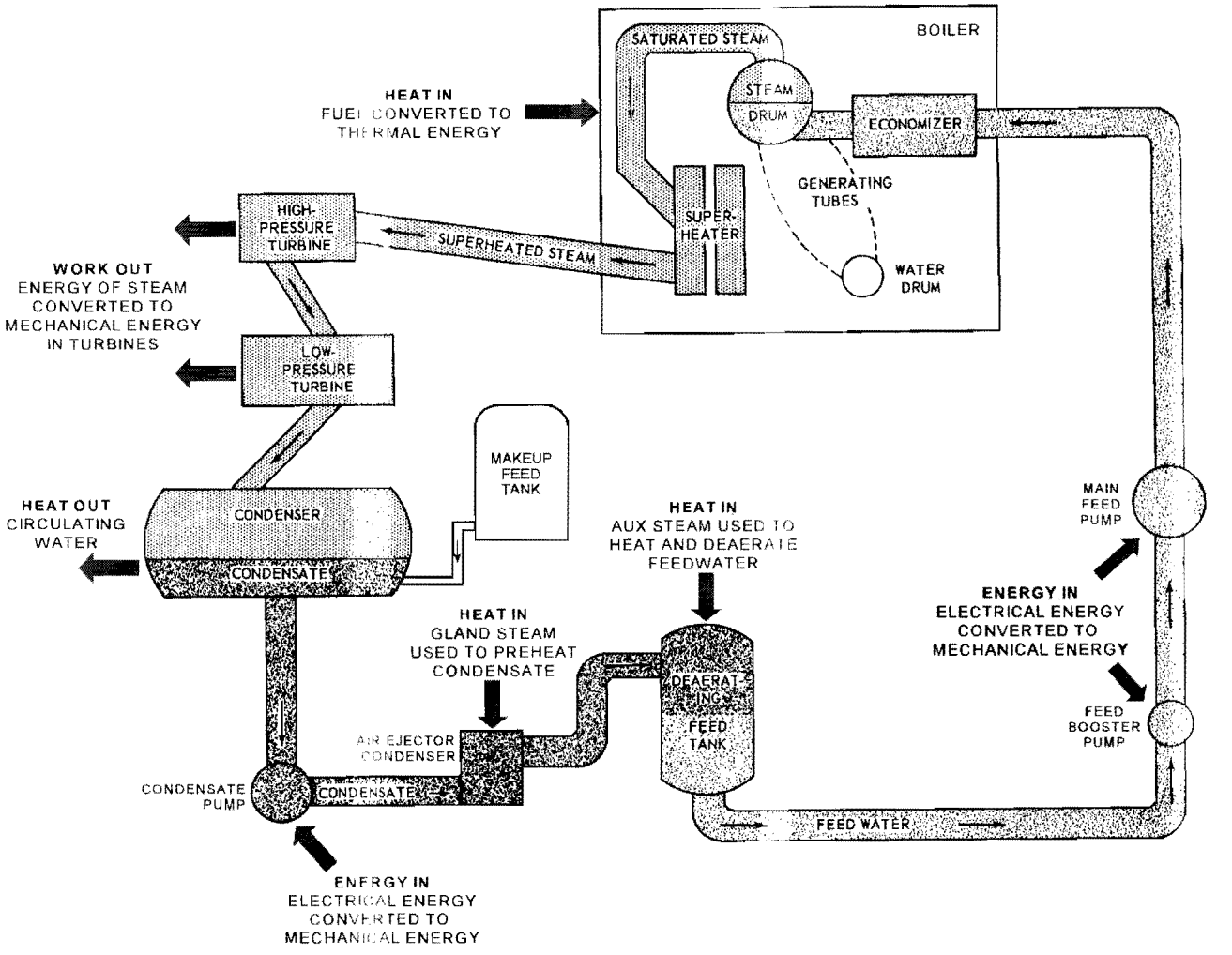

Figure 10 - Energy Relationships in the Basic Power Plant Cycle

8.2 Energy Balances

From previous discussion, it should be apparent that putting 1 Btu in at the boiler furnace does not mean that 778 foot-pounds of work will be available for the production of electricity. Some of the energy put in at the boiler is used by auxiliary machinery such as pumps and forced draft blowers to supply the boiler with feedwater, fuel, and combustion air. In addition, there are many 'energy losses' throughout the power plant which greatly affect its overall efficiency.

As previously discussed, energy cannot actually be 'lost'. But when it is transformed into a form of energy which we cannot use, it is common to say there has been an energy loss. Since no insulation is perfect (100% efficient), some thermal energy is always lost as steam travels through piping. In addition, friction losses occur in all machinery and piping; some heat must be wasted as the combustion gases go up the stack; and some heat must be lost at the condenser as the steam exhausted from the turbines gives up heat to the circulating water. Therefore, it cannot be expect that all of the fuel and heat supplied to a system will be converted into work. Even in the most efficient possible cycle, we know that some heat must always be rejected to a receiver which is at a lower temperature than the source. Thus, each Btu that is theoretically put in at the boiler must be divided up in a number of ways before the energy can be completely accounted for. However, the energy account will always balance; *energy in will always equal energy out.*

Designers of engineering equipment use energy balances to analyze energy exchanges and to compute the energy requirements for proposed equipment or plants. Operating engineers use energy balances to evaluate plant performance. Power plant engineers may refer to energy balance diagrams in order to find out whether the plant is operating at designed efficiency or whether defects are causing unnecessary waste of steam, fuel, and energy.

An energy balance for an entire power plant is usually made up in the form of a flow diagram similar to (but more detailed than) the one shown in Figure 10. A number of numerical values are entered on the flow diagram, the most important of which are the quantities of the working fluid flowing per hour at various points and the thermodynamic states of the working fluid at various points. The quantity of fluid flowing per hour may be obtained by direct measurement of flow through flow meters or nozzles or by calculation. In some instances, it is necessary to estimate steam consumption of pumps and other units on the basis of available test data. Data on the state of the working fluid is obtained from pressure and temperature readings. Enthalpy calculations are made and noted at various points on the diagram. The complete energy balance includes tabular data as well as the data shown on the flow diagram.